

Modeling of Viscoelastic Contacts and Evolution of Limit Surface for Robotic Contact Interface

Paolo Tiezzi and Imin Kao, *Member, IEEE*

Abstract—Viscoelastic contact is a type of contact which includes, in addition to linear or nonlinear elastic response, time-dependent response due to relaxation or creep phenomena that govern the contact behavior. The characteristics of the time-dependent *relaxation* of such a viscoelastic contact are typically exponentially decaying functions, and exponentially growing functions for *creep*, respectively. Such contacts can be found in anthropomorphic robotic fingers, soft materials, viscoelastic skin with rigid core, and human fingers and feet. In this paper, the nature of viscoelastic contacts is investigated, and the evolution of their friction limit surfaces and of the pressure distributions at the contact interface are studied. Two cases commonly found in robotic grasping and manipulation are discussed. Based on the modeling formulation, it is found that the two important parameters of analysis and modeling for such contacts, i.e., the radius of contact area and the profile of pressure distribution, can be chosen using proposed coupling equations as the viscoelastic contact interface evolves with time. The new contribution of this paper includes a proposal of coupling equations between the two important parameters to describe the viscoelastic contact interface, and a study of the evolution of limit surfaces for viscoelastic contact interface due to temporal dependency, and the implication on grasp stability. It is found from the evolution of limit surfaces that when normal force is applied with typical viscoelastic contacts, grasp becomes more stable as time elapses. The modeling can be applied to the design of fingertips and the analysis of robotic grasping and manipulation involving viscoelastic fingers.

Index Terms—Contact mechanics, creep, evolution of friction limit surface, relaxation, viscoelastic finger.

I. INTRODUCTION

DIFFERENT types of contacts have been proposed for analysis and modeling contacts in robotic grasping and manipulation [1]. The general framework of “*limit surface*” has been developed to capture a conceptual 3-D surface within which the contact, with applicable contact interface accompanied by the applied forces and moments, will sustain without slip [2]–[6]. The limit surface has become a very useful tool for modeling contact interface in robotic grasping and manipulation.

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P. Tiezzi is with the Department of Mechanical Engineering (DIEM), University of Bologna, 40136 Bologna, Italy (e-mail: paolo.tiezzi@mail.ing.unibo.it).

I. Kao is with the Department of Mechanical Engineering, SUNY at Stony Brook, Stony Brook, NY 11794-2300 USA (e-mail: kao@mal.eng.sunysb.edu).

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Point contact with or without friction was first employed to model contacts in robotics [7]. In the case of point contact with friction, the well-known Coulomb’s law was often employed [8], [9]. As a result, the friction limit surface is a *friction cone* with the half of the conical included angle being $\theta = \tan^{-1} \mu$, where μ is the coefficient of friction at the contact interface.

However, when the contact patch assumes a finite area, the assumption of point contact is no longer valid, and needs to be extended to *soft contact*, which includes not only a friction (or traction) force at the contact interface, but also a moment sustained due to the finite area of contact [2], [4]–[6], [10], [11]. It has been shown that the limit surface of soft-finger contact resembles an ellipsoid with friction force and moment in the major and minor axes, respectively [3], [4], [6]. As the materials and geometric designs of fingertips diversify, the viscoelastic behavior of certain types of fingertips was observed, especially with the relaxation of fingertip contact force or the creep of contact area, characteristics of viscoelastic contact [12]. The modeling of such contact finds applications in modeling of human and biomedical fingertips (for example, in [13] and [14]).

In this paper, we study the characteristics of contact interface and limit surface for viscoelastic contact interface which can be applied to robotic fingers in grasping and manipulation, as well as robotic feet. Two different cases of modeling with either: 1) maintaining constant contact area (due to a constant normal displacement); or 2) maintaining constant normal force at the contact interface are presented and discussed, in conjunction with the time-dependent nature of viscoelastic contacts. The new contribution of this paper includes the incorporation of the time-dependent nature of viscoelastic contact interface into the modeling of grasping and manipulation. Specifically, we proposed two coupling equations to correlate the two important parameters of contact modeling (the radius of contact area and the profile of pressure distribution across the contact interface), and studied the evolution of limit surfaces for viscoelastic contact interface, due to its time-dependent nature, as well as the implication of such evolving limit surfaces on the stability of grasping and manipulation.

The following sections are organized as follows. In Section II, we present the formulation of modeling for both elastic and time-dependent viscoelastic features. In Sections III and IV, we discuss two cases common to grasping and manipulation using viscoelastic contacts. Practical choices and correlation between the time-dependent contact area and pressure distribution are proposed. Section V presents the construction of limit surfaces and results, followed by discussions and conclusion in Sections VI and VII, respectively.

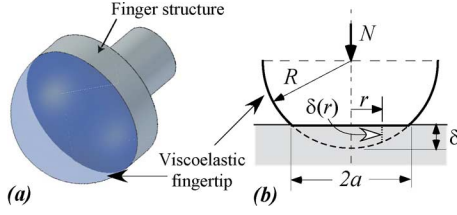


Fig. 1. (a) Hemispherical viscoelastic fingertip, with a radius of R . (b) The fingertip pressing against a flat, rigid surface is shown with definition of terminology. The diameter of the circular contact area is $2a$, and the maximum normal displacement or depression is δ at the center of contact patch. The depression of fingertip surface upon the surface at a distance r from the center is $\delta(r)$. The normal force applied is N , as shown.

II. MODELING OF VISCOELASTIC FINGERTIPS

A hemispherical viscoelastic fingertip is shown in Fig. 1(a). The fingertip makes contact with a rigid flat surface by the application of a normal force N , with corresponding normal moment and tangential force at the contact interface. For the sake of modeling, the contact is assumed to be sustained without slip; that is, the resulting tangential force and normal moment at the contact interface are within the limit surface. Fig. 1(b) shows a schematic of such contact with parameters used to describe the contact interface. The fingertip can be, in general, solid and homogeneous, as discussed in [6], or its structure can have a constant thick, soft layer covering a rigid “core,” as investigated in [15] and [16].

From Fig. 1, the following geometrical relationship can be formulated:

$$a^2 = R^2 - (R - \delta)^2. \quad (1)$$

Furthermore, the following general equation of pressure distribution is adopted [5], [6]:

$$p(r) = C_k \frac{N}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} \quad (2)$$

where C_k is a coefficient, a function of k , that adjusts the profile of pressure distribution to satisfy the equilibrium condition at the contact interface. In his paper in 1882 [17], Hertz first proposed a pressure distribution, corresponding to $k = 2$ in (2), for linear elastic contact with small deformation, which was adopted later, for example, in [12].¹ However, the value for k is not necessarily 2 for a general contact pressure distribution. In fact, in a typical viscoelastic contact, after relaxation due to the time-dependent characteristics, the contact pressure will be more uniformly distributed, corresponding to a higher k value in (2). Other formulations of pressure distribution for contacts of nonlinear solids are reported in [12] and, in particular, Matthews [18] proposed an expression of the contact of spheres which, similar to (2), considers a pressure distribution at the contact interface different from a parabolic² profile as in [17].

¹This equation with $k = 2$, in its elegant form, can be integrated with a closed-form solution, which might be one of the reasons why Hertz chose this type of equation for pressure distribution.

²Matthews [18] proposed and experimentally validated the equation $p(r) = p_m(2n+1)/2n[1 - (r/a)^2]^{-(1/2)n}$, where n ($n \geq 1$) is the stress-hardening exponent of the equation $\epsilon = \epsilon_0(\sigma/\sigma_0)^n$ in a uniaxial tensile test.

We adopt (2) and allow k to vary to render different pressure distributions across the contact surface, as illustrated in Fig. 2. Moreover, the parameters a and N in (2) are not necessarily constants, due to the relaxation and creep of viscoelastic contacts. Therefore, it is important to note the general formulation of pressure distribution, and to recognize that the pressure distribution for viscoelastic fingers is a function of time.

The coefficient C_k can be obtained by integrating (2) over the entire contact area A , and imposing the equilibrium condition at the contact interface

$$N = \int_A p(r) dA = \int_0^{2\pi} \int_0^a p(r) \cdot r dr d\theta \quad (3)$$

from which it is derived that [6]

$$C_k = \frac{3}{2} \frac{k \Gamma(\frac{3}{k})}{\Gamma(\frac{1}{k}) \Gamma(\frac{2}{k})} \quad (4)$$

where Γ is the Gamma function, and k is a positive real number, although integer values are often used. It is apparent from (4) that C_k is only a function of k .³

Due to the nature of the time-dependent functions for the modeling of contact interface of viscoelastic fingers, two different cases are discussed in Sections III and IV. In the first case, a prescribed constant normal displacement (resulting in constant contact area) is considered and the influence of relaxation is investigated, as illustrated in Fig. 3(a). The second case considers the creep phenomenon due to a constant normal load, as shown in Fig. 3(b).

In order to describe the viscoelastic behavior, a general approach [19] considers the force and the displacement related by functions in the form of

$$N(t) = \mathcal{G}(\delta, t) \quad (5)$$

$$\delta(t) = \mathcal{H}(N, t) \quad (6)$$

that can be very complex to exploit. The function $\mathcal{G}$ is named “relaxation function,” while the function $\mathcal{H}$ is the “creep compliance.” In the general case of nonlinear viscoelasticity, the relaxation function $\mathcal{G}$ specifies the force response, as time elapses, to a step displacement δ from the undeformed configuration, while the creep compliance $\mathcal{H}$ gives the displacement response to a step force N from the undeformed configuration. Because creep and relaxation are two aspects of the same viscoelastic phenomena, the two functions are related [19].

When the linear hypothesis holds, the functions $\mathcal{G}$ and $\mathcal{H}$ become only function of time and specify, respectively, the force response to a unit step change in displacement and the displacement response to a unit step change in force. In this particular case, (5) and (6) become

$$N(t) = \hat{\mathcal{G}}(t) \cdot \delta \quad (7)$$

$$\delta(t) = \hat{\mathcal{H}}(t) \cdot N \quad (8)$$

where δ is the imposed displacement at the contact, and N is the normal force.

³Equation (4) can also be expressed in an alternative form with $C_k = (3\Gamma(3/k))/\Gamma(1/k)\Gamma(1+2/k)$ because $\Gamma(z+1) = z\Gamma(z)$.

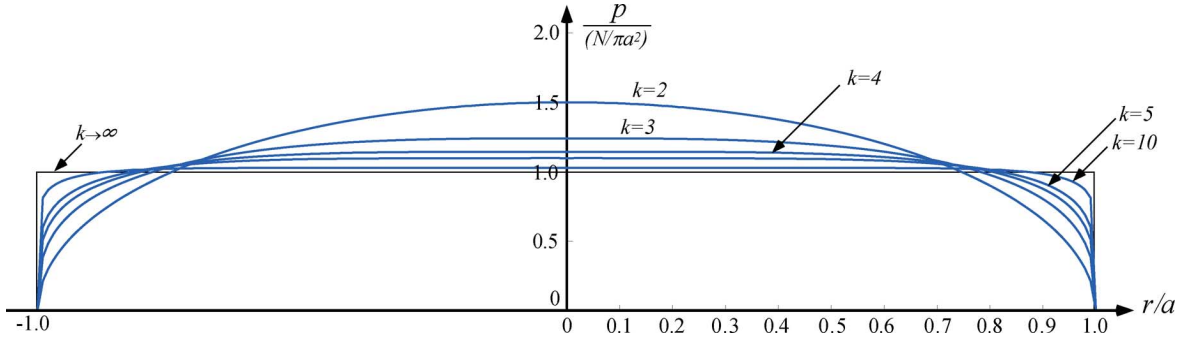


Fig. 2. Pressure distribution of viscoelastic contact based on (2) for different values of k . As $k \rightarrow \infty$, the distribution becomes uniform with normalized maximum pressure of 1.0. Selected values of C_k are tabulated in Table I.

TABLE I
TABLE OF VALUES OF C_k FOR SELECTED VALUES OF k . THE VALUES OF k CAN BE INTEGER OR REAL NUMBERS, DEPENDING ON THE CHARACTERISTICS OF THE MATERIALS AND GEOMETRY OF CONTACT

k	C_k
1.8	1.6080
2	1.5000
3	1.2405
4	1.1441
10	1.0274
100	1.00032

In order to overcome the difficulties of the formulation (5) and (6) in the general case of nonlinear viscoelasticity, the hypothesis of Fung [20] will be adopted in this paper, and the “reduced relaxation function” $g(t)$ and the “reduced creep compliance” $h(t)$ will be introduced in the following sections in order to describe the time-dependent behavior of viscoelastic contacts.

Before we move on, it is worth noting that the hypothesis of Fung [20] segregates the elastic response from the temporal response. This approach enables us to employ elastic response based on various models developed for soft fingers, independent of the temporal response for viscoelastic contacts. Essentially, the viscoelastic contact modeling can be thought of as the concatenation of the elastic (linear or nonlinear) response, a function of the imposed δ or N , and the temporal response (with relaxation or creep), a function of time t .

III. CONTACT WITH IMPOSED DISPLACEMENT

For the first case illustrated in Fig. 3(a), we consider the hemispherical viscoelastic fingertip pressing onto the contact surface while maintaining a constant step displacement δ_0 after the contact is made. Due to the viscoelastic behavior, relaxation phenomenon occurs and the normal load will decrease over time [12], [19], as illustrated in Fig. 3(a).

Because the normal displacement is held constant, the radius of contact area a also remains constant due to the contact geometry, as given by (1). The pressure distribution over the entire contact area, due to the fact that the total normal force varies over time, changes according to the following equation, modified from (2):

$$p(r, t) = C_k \frac{N(t)}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} \quad (9)$$

where the normal force $N(t)$ becomes time-dependent, while the parameters a , k , and C_k remain constant as in (2), because the contact displacement and radius do not change. The shape of the pressure distribution is also assumed to be constant.

To model the relaxation of normal force N after the contact is made, and in order to overcome the difficulties of formulation in (5), we apply the noteworthy model proposed by Fung [20] for the human tissues and adopted by Pawluk and Howe [13], and Jindrich *et al.* [21] to model the human fingertip behavior. Fung assumed that the *relaxation function* ($\mathcal{G}$) has the form of

$$\mathcal{G}(\delta, t) = N^{(e)}(\delta) \cdot g(t) \quad \text{with} \quad g(0) = 1 \quad (10)$$

where $N^{(e)}(\delta)$ is the *elastic response*, with superscript “(e)” denoting the elastic response, and $g(t)$ is the *reduced relaxation function* which describes the time-dependent behavior of the material. Under these assumptions, a quasi-linear viscoelastic model has been developed [22], [23].

The term $N^{(e)}(\delta)$ is the amplitude of the force generated instantaneously by a displacement δ from the undeformed configuration. The nonlinear elastic response $N^{(e)}$ can be modeled through different analytical expressions. Two important models of the elastic stiffness $K^{(e)} = dN/d\delta$ adopted in the literature are

$$K^{(e)}(\delta) = m \cdot e^{b\delta} \quad (11)$$

$$K^{(e)}(\delta) = p \cdot \delta^q \quad (12)$$

where (m, b) and (p, q) are parameters which depend on the materials and geometry. The expression of $N^{(e)}(\delta)$ can be obtained from the integration of (11) or (12) with respect to δ , as well as the initial condition $N^{(e)}|_{\delta=0} = 0$

$$N^{(e)}(\delta) = \frac{m}{b} (e^{b\delta} - 1) \quad (13)$$

$$N^{(e)}(\delta) = \frac{p}{q+1} \delta^{q+1}. \quad (14)$$

Pawluk and Howe [13] and Barbagli *et al.* [24], exploited (11) to model the relationship between normal force and normal displacement in human finger indentation. Han and Kawamura [25] compared the human finger stiffness with that of artificial fingers using both (11) and (12). Kao and Yang [26], starting from previous research results [6], derived an expression for nonlinear stiffness of soft contact that can be associated with (12). Tiezzi

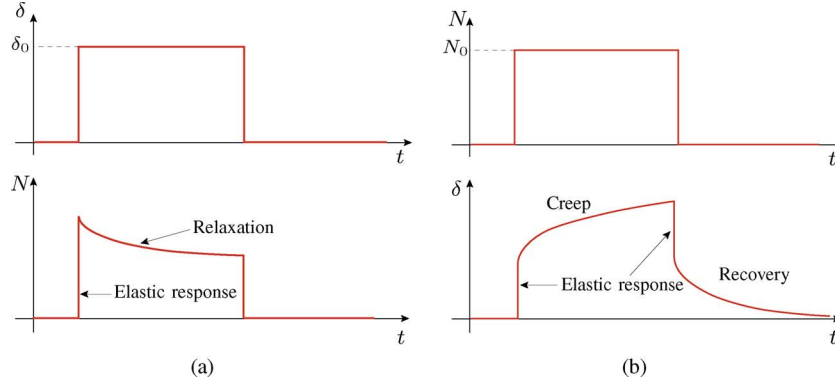


Fig. 3. (a) Case I: schematic of the relaxation of normal force N while maintaining a constant step displacement δ_0 , and radius of contact a . (b) Case II: schematic of the creep of normal displacement δ as a function of time, while maintaining a constant normal force N_0 . A square function of normal load is illustrated in order to point out the recovery phase after the removal of the load.

and Vassura [15] exploited both (11) and (12) to investigate the behavior of elastic skins covering a rigid fingertip structure.

The reduced relaxation function $g(t)$ is a time-decaying function. When normalized to 1 at $t = 0$, it can be expressed by the following form [13], [19]:

$$g(t) = \sum_{i=0}^n c_i e^{-\nu_i t} \quad \text{with} \quad \sum_{i=0}^n c_i = 1 \quad \text{and} \quad \nu_0 = 0 \quad (15)$$

where the parameters c_i and ν_i depend on the material of the viscoelastic interface, and the exponents ν_i identify the rates of the relaxation phenomena.

The relaxation function $\mathcal{G}$, introduced in (10), defines the force response to a step displacement δ from the undeformed configuration as time elapses. Therefore, in the case of a single step displacement $\delta = \delta_0$, the force response will be

$$N(t) = N^{(e)}(\delta_0) \cdot g(t). \quad (16)$$

The pressure distribution at the contact interface of the viscoelastic finger is obtained by substituting (15) and (16) into (9) to render

$$p(r, t) = \frac{C_k}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} \left[c_0 + \sum_{i=1}^n c_i e^{-\nu_i (t-\tau)} \right] N^{(e)}(\delta_0). \quad (17)$$

The first group in (17) depends only on r and does not depend on time. On the contrary, the second term expresses the time dependence of the pressure distribution due to the changing normal force $N(t)$. Fig. 4 shows the calculated results of the evolution of the pressure distribution due to the relaxation of the normal load in the case of a constant normal displacement δ_0 with a second-order pressure distribution profile ($k = 2$). Fig. 4(a) and (b) plot the normalized force N/N_0 and pressure p/p_0 , respectively, in order to highlight only the relaxation

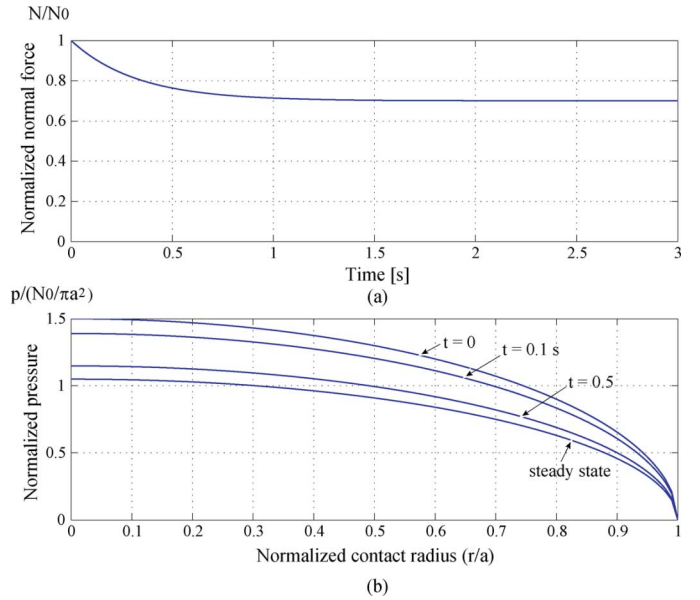


Fig. 4. (a) Relaxation of normal load when a constant normal displacement is imposed. The normal load $N(t)$ is normalized with respect to the elastic response $N_0 = N^{(e)}(\delta_0)$. (b) Evolution of the pressure distribution due to the relaxation of the normal load. The pressure is normalized with respect to the mean value at the initial instant $p_0 = p_m|_{t=0} = N_0/(\pi a^2)$. A second-order profile ($k = 2$ and $C_k = 1.5$) of the pressure distribution is considered, while the parameters of the reduced relaxation function (15) are deduced from the experimental results described in [22], [23]: $c_0 = 0.7$, $c_1 = 0.3$, $\nu_1 = 3.1$.

phenomenon. The parameters of the reduced relaxation function $g(t)$, used to plot the graphics, are $n = 1$, $c_0 = 0.7$, $c_1 = 0.3$, $\nu_1 = 3.1$, which are derived from the experimental results reported in [22] and [23].

If the imposed displacement δ is varying, the contribution of the whole past history must be considered. The force produced by an infinitesimal displacement $d\delta(\tau)$, superposed in a state of displacement δ at an instant of time τ , with $t > \tau$, is

$$dN(t) = g(t - \tau) \frac{dN^{(e)}(\delta(\tau))}{d\delta} d\delta(\tau) \quad (18)$$

as deduced from (16). By applying a modified superposition principle [19], [27], [28], the total force at the time instant t is the sum of contribution of all the past changes; that is

$$N(t) = \int_0^t g(t-\tau) \frac{dN^{(e)}(\delta(\tau))}{d\delta} \frac{d\delta(\tau)}{d\tau} d\tau. \quad (19)$$

Equation (19) can be rewritten in the following form:

$$N(t) = \int_0^t g(t-\tau) \cdot K^{(e)}(\delta(\tau)) \cdot \dot{\delta}(\tau) d\tau \quad (20)$$

where $K^{(e)}(\delta)$ is the elastic stiffness as expressed by (11) or (12), and $\dot{\delta}(\tau)$ is the rate of displacement.

The evolution of the pressure distribution is obtained by substituting (15) and (20) into (9). That is

$$p(r, t) = \frac{C_k}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} \int_0^t \left[c_0 + \sum_{i=1}^n c_i e^{-\nu_i \cdot (t-\tau)} \right] K^{(e)}(\delta) \dot{\delta}(\tau) d\tau. \quad (21)$$

IV. CONTACT WITH IMPOSED NORMAL LOAD

For the case in which the normal load ($N = N_0$) applied on the hemispherical fingertip is maintained at constant, the normal displacement δ will increase over time due to the nature of viscoelastic creep phenomena [19], as shown in Fig. 3(b). Furthermore, it is well known that by keeping the normal force constant, the pressure distribution of viscoelastic contact will gradually become more uniform due to relaxation [12], [29]. In fact, as a result of the increasing normal displacement, the radius of contact area a also increases. However, the equilibrium condition at the contact interface imposes a constraint equation, as in (3), with the normal force N being constant.

Because the contact area increases while the normal force is held constant, the shape of pressure distribution also changes, implying the changes in the exponent k in (2). For a given N and a , the pressure distributions for various typical k values are plotted in Fig. 2.

In the following modified equation (22) of the pressure distribution, both a and k are function of the time

$$p(r, t) = C_{k(t)} \frac{N}{\pi a^2(t)} \left[1 - \left(\frac{r}{a(t)} \right)^{k(t)} \right]^{1/k(t)} \quad (22)$$

where $C_{k(t)}$ denotes the coefficient C_k as a function of $k(t)$. The maximum pressure takes place at $r = 0$, and is given by

$$p_{\max}(t) = p(0, t) = C_{k(t)} \frac{N}{\pi a^2(t)}. \quad (23)$$

The maximum pressure given by (23) decreases while the contact area increases. In Fig. 2, the normalized maximum pressure at $r = 0$ corresponds to the coefficient C_k at each value of k . The total area under each half-curve is unity, as indicated in (24). Note that as $k \rightarrow \infty$, the pressure becomes uniformly

distributed, as expected. Substituting (22) into (3), the integral constrain equation becomes

$$\frac{2C_{k(t)}}{a^2(t)} \int_0^{a(t)} r \left[1 - \left(\frac{r}{a(t)} \right)^{k(t)} \right]^{1/k(t)} dr = 1. \quad (24)$$

A. Contact Area and Pressure Distribution

The constraint equation in (24), when integrated, will yield (4), which does not contain $a(t)$ at any given instant t . This is owing to the specific form of pressure distribution employed in (2). This has profound implication on the analysis of contact and the behavior of viscoelastic fingers. We note the following observations.

- Theoretically, the exponent k can be chosen independent of the constraint equation in (24). The choice of k is primarily determined by the shape of the pressure distribution at the contact interface. Larger value of k indicates more uniform pressure distribution. A typical range of values of $k = 2 \sim 4$ can be used.
- The constraint (24), when integrated, does not yield an equation which retains the radius of contact a . Therefore, the parameter a is independent of the equilibrium condition expressed by (24).
- Due to the specific form of the pressure distribution, the equilibrium condition (24) imposes a value of C_k , expressed by (4), that relates the maximum pressure $p_{\max} = p|_{r=0}$ and the mean pressure $N/\pi a^2$, taking into account the pressure distribution due to k .

Although the two important parameters, radius of contact a and profile of pressure distribution k , are seemingly independent based on the theoretical modeling as proposed herein, it is postulated that the two parameters k and a are correlated, according to the properties of the fingertip material as well as the geometric configuration of the fingertip. In order to facilitate the analysis and formulation for the modeling of viscoelastic contact interface, it is advantageous to derive a coupling equation for the two important parameters. Such a coupling equation needs to satisfy the equation equilibrium and physical coherent behavior. In the following sections, two such coupling equations are proposed and discussed.

B. Coupling Equation Based on Pressure Distribution

In this section, we derive a coupling equation based on the evolution of the profile of pressure distribution at the contact interface. In order to determine the parameters a and k , we recognize the fact that as the relaxation is taking place, the pressure distribution becomes more uniform with gradually increasing k , as illustrated in Fig. 2, and the radius of contact area increases too [29]. We also assume that the exponent k varies from 2 to 4, as relaxation takes place. This choice is consistent with the physical behavior one can expect. Moreover, it corresponds to the values of k obtained in Section IV-C.

Next, we choose the radius of contact a in such a way that

$$a \propto \frac{1}{\sqrt{C_k}}. \quad (25)$$

TABLE II
SAMPLE VALUES OF C_k AND a/a_0 CORRESPONDING TO DIFFERENT
VALUES OF THE PARAMETER k BETWEEN 2 AND 4

k	C_k	$1/\sqrt{C_k}$ in Eq. (64)	a/a_0 from Eq. (65)
2	1.50	0.822	1.0
2.5	1.333	0.864	1.051
3	1.241	0.896	1.090
3.5	1.183	0.920	1.120
4	1.144	0.939	1.143

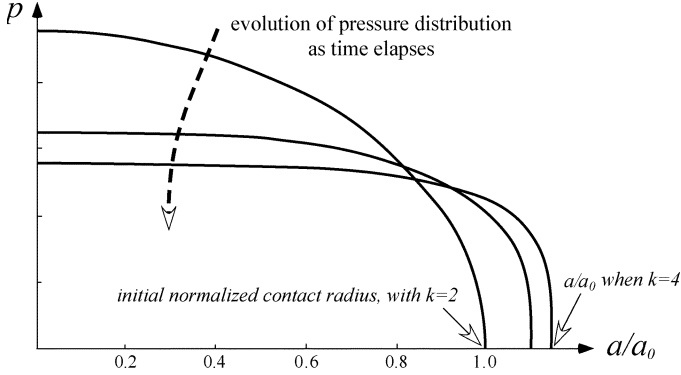


Fig. 5. Evolution of pressure distribution when the normal contact force is maintained at constant, resulting in the relaxation and growth of the contact radius, resembling exponential growth from a_0 to a_f as a function of time. As $t \rightarrow \infty$ when the pressure profile is corresponding to $k = 4$, the pressure distribution approaches a constant profile. As time elapses, the parameter k increases, corresponding to the flattening of the pressure distribution as relaxation takes place for the viscoelastic finger.

As we stated in Section IV-A, the coupling equation needs to satisfy the equilibrium equation and physical coherent behavior. It is important to note that the assumption of relationship in (25) satisfies such criteria. Furthermore, the postulation in (25) gives a growing contact radius a when the value of C_k decreases due to a more uniform pressure distribution. Therefore, this relationship is consistent with the well-known physical behavior.

Based on the choice of the contact radius a described in (25), we can obtain a weighted least-squares (LS) best fit of the form in (59) to describe the normalized contact radius (the details are in Appendix II)

$$\frac{a}{a_0} \cong c(1 - 0.52e^{-0.535k}) \quad (26)$$

where a_0 is the radius of contact at $t = 0$ when $k = 2$, and $c = 1.217$, a constant. Equation (26) encompasses changes in pressure distribution corresponding to $2 \leq k \leq 4$ as time changes from $0 \leq t < \infty$. Table II lists a few values of k between 2 and 4 with corresponding values of growing normalized radius a/a_0 .

The evolution of contact pressures and areas as time elapses is obtained using (26) and illustrated in Fig. 5, where the relaxation of the viscoelastic finger with constant normal force results in the flattening of the pressure distribution at the contact, as well as the enlargement of contact radius. The results are consistent with the physical behavior of viscoelastic contacts [29].

C. Coupling Equation Based on Creep Compliance

In order to obtain a relationship between the parameters a and k , an alternative approach is presented in this section based

on the creep compliance. A constitutive equation is introduced to relate both parameters with the viscoelastic property of the fingertip material. Adopting the same modeling approach as that in Section III, the creep compliance $\mathcal{H}(N, t)$ is assumed to be in the form of

$$\mathcal{H}(t, N) = \delta^{(e)}(N) \cdot h(t) \quad \text{with} \quad h(0) = 1 \quad (27)$$

where the function $h(t)$ is the *reduced creep compliance*, which describes the time-dependent behavior of the fingertip, and $\delta^{(e)}(N)$ denotes the *elastic response*, which is the instantaneous displacement generated by the force N from the undeformed configuration.

The expressions of the *elastic displacement* $\delta^{(e)}(N)$ are obtained by inverting (13) and (14), which specify the elastic force response $N^{(e)}(\delta)$

$$\delta^{(e)}(N) = \frac{1}{b} \ln \left(\frac{b}{m} N + 1 \right) \quad (28)$$

$$\delta^{(e)}(N) = \left(\frac{q+1}{p} N \right)^{1/(q+1)}. \quad (29)$$

The *reduced creep compliance* can be expressed, without loss of generality [19], by the following expression:

$$h(t) = 1 + \sum_{i=1}^n c_i (1 - e^{-\eta_i t}) \quad (30)$$

where the parameters c_i and η_i are constants depending on the materials, and the exponents η_i identify the rates of creep phenomena.

In case a single step of normal load ($N = N_0$) is applied at the initial instant $t = 0$, the displacement response $\delta(t)$ is given by the creep compliance $\mathcal{H}(t, N)$, as in (27); that is

$$\delta(t) = \delta^{(e)}(N_0) \cdot h(t). \quad (31)$$

As a result of the increase of the normal displacement, the radius of contact area increases, too. For the sake of simplicity, the relationship between the radius of the circular contact area a and the displacement δ can be derived from (1) as

$$a^2 \cong 2R\delta \quad (32)$$

by neglecting the second-order term in δ^2 . The radius of contact obtained from the approximation in (32) is slightly larger than that from (1).⁴ Nevertheless, the radius of contact given by (32) is closer to the actual contact radius that is affected by the expansion due to the conservation of the volume of the pad at the contact interface.

Substituting (31) into (32), we obtain

$$a^2(t) \cong 2R\delta(t) = 2R\delta^{(e)}(N_0) \cdot h(t) \quad (33)$$

⁴The difference is the second-order term δ^2 . For example, the discrepancy is about 10% when $\delta/R = 20\%$, corresponding to quite a large depression at the contact.

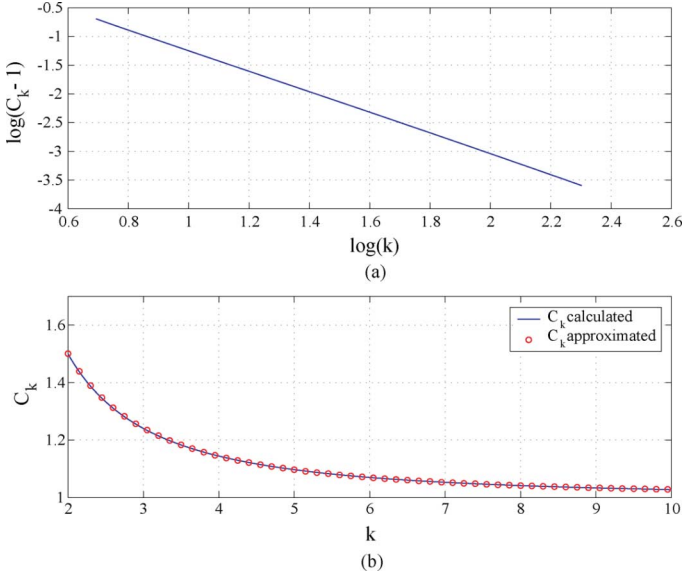


Fig. 6. (a) Linear relationship between $\log(k)$ and $\log(C_k - 1)$ is plotted. (b) Comparison between the calculated and approximated C_k values in (38), as a function of k shows very good match of the results.

that can be rewritten as

$$a(t) \cong a_0 \cdot \sqrt{h(t)} \quad (34)$$

where $a_0 = 2R\delta^{(e)}(N_0)$ is the contact radius at the initial instant $t = 0$.

A useful hypothesis, proposed in (25), is $C_k \propto 1/a^2$. Combining it with (34), we have

$$C_k \propto \frac{1}{a^2} \propto \frac{1}{h(t)}. \quad (35)$$

Thus, it is possible to relate the coefficient C_k and the reduced creep compliance h as follows:

$$C_k = \frac{3}{2} \frac{k\Gamma(\frac{3}{k})}{\Gamma(\frac{1}{k})\Gamma(\frac{2}{k})} = \frac{C_{k0}}{h(t)} \quad (36)$$

where $C_{k0} = C_k(k|_{t=0})$ is adopted because $h|_{t=0} = 1$. For example, when the initial pressure profile is second-order, i.e., $k|_{t=0} = 2$, the corresponding value for C_{k0} is 1.5.

From (36), the relationship between k and the time can be obtained. The direct use of (36) is not convenient, because the variable k is embedded in the arguments of the Gamma function. Therefore, an equivalent relationship between C_k and k needs to be obtained.

It can be observed that the two quantities $\log(C_k - 1)$ and $\log(k)$ are linearly related [see Fig. 6(a)]. By means of the LS best fit, the following approximate relationship can be obtained:

$$\log(C_k - 1) = a_1 \log(k) + a_2 \quad (37)$$

with $a_1 = -1.7984$ and $a_2 = 0.5545$. After some mathematical manipulation, it can be derived that the approximate relationship between C_k and k can be expressed as

$$C_k \cong e^{a_2} \cdot e^{a_1 \log(k)} + 1. \quad (38)$$

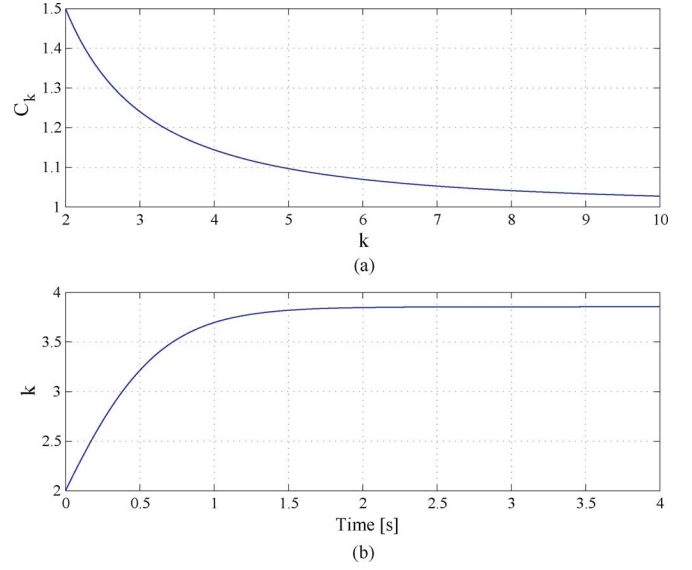


Fig. 7. (a) Plot of C_k as a function of k , as expressed by (4). (b) Plot of k as a function of time, as expressed by (40), by adopting a reduced creep compliance $h(t)$, as expressed by (41).

Equation (38) is a very good approximation of $C_k(k)$ over a larger range of k [see Fig. 6(b)], where the constants a_1 and a_2 are obtained through the LS method.

Substituting (38) into (36), we derive

$$\alpha_2 \cdot e^{\alpha_1 \log(k)} + 1 = \frac{C_{k0}}{h(t)} \quad (39)$$

where $\alpha_1 = a_1$ and $\alpha_2 = e^{a_2}$. Thus, the change of k as a function of time can be obtained from (39) as follows:

$$k(t) = \left[\frac{\frac{C_{k0}}{h(t)} - 1}{\alpha} \right]^\beta = \left[\frac{\frac{C_{k0}}{h(t)} - 1}{1.741} \right]^{-0.556} \quad (40)$$

where $\alpha = \alpha_2 = 1.741$ and $\beta = \alpha_1^{-1} = -0.556$. Equation (40) shows that the relationship between k and time is only affected by the properties of the material through the *reduced creep compliance* $h(t)$. Fig. 7 shows the plots of C_k as a function of k , as expressed by (4), and k as a function of time, as expressed by (40). The function $h(t)$ is assumed to be in the form of (30) with only two parameters; that is

$$h(t) = 1 + c_1(1 - e^{-\eta_1 t}) \quad (41)$$

where $c_1 = 0.3$ and $\eta_1 = 3.1$. This choice is consistent with the experimental results obtained in [22] and [23].

Substituting (34), (40), and (41) into (22), the pressure distribution $p(r, t)$ can be obtained. Fig. 8 illustrates the results of the creep of the viscoelastic finger by keeping the normal force constant, resulting in the flattening of the pressure distribution at the contact, as well as the enlargement of the contact radius. Due to the choice of the reduced creep compliance (41), the resulting relaxation and growth of the contact area are exponential. As time elapses, the pressure distribution approaches a more constant profile, with k varying from 2 to 4 or higher [see Fig. 7(b)] and the radius of the contact area increases by approximately 15%.

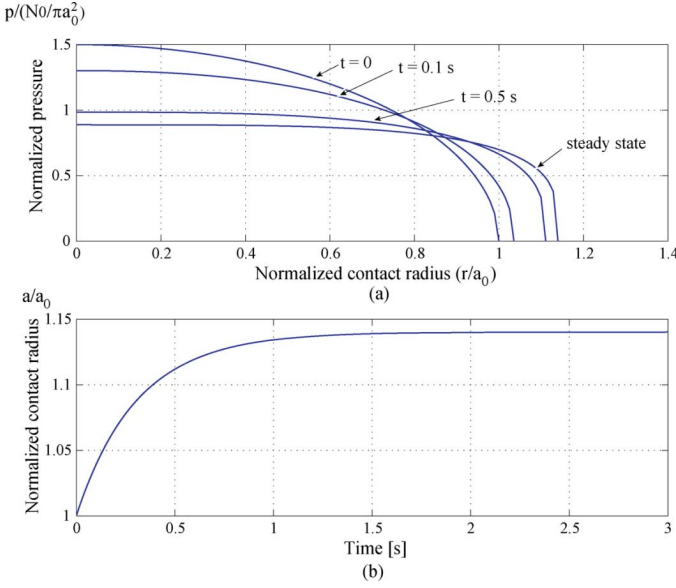


Fig. 8. (a) Plot shows the evolution of the pressure distribution as a function of the normalized contact radius r/a_0 , when a single step of normal contact force $N = N_0$ is applied and held at constant. The pressure is normalized with respect to the mean value at the initial instant $p_0 = p_m|_{t=0} = N_0/(\pi a_0^2)$, where a_0 is the radius of contact at $t = 0$. (b) The change of the contact radius as a function of time. The parameters of the reduced creep compliance in (30) are chosen coherently with the experimental results obtained in [22] and [23]: $n = 1$, $c_1 = 0.3$, $\eta_1 = 3.1$.

In the general case, when the normal load N is not constant but varying, the contribution of the whole past history must be considered. Similar to Section III, the dual relationship of (19) can be derived by substituting $N^{(e)}(\delta)$ with $\delta^{(e)}(N)$ and $g(t)$ with $h(t)$. Therefore, the change of the normal displacement δ as a function of time can be expressed as

$$\delta(t) = \int_0^t h(t-\tau) \frac{d\delta^{(e)}(N(\tau))}{dN} \frac{dN(\tau)}{d\tau} d\tau \quad (42)$$

that can be rewritten as

$$\delta(t) = \int_0^t h(t-\tau) \cdot C^{(e)}(N(\tau)) \cdot \dot{N}(\tau) d\tau \quad (43)$$

where $C^{(e)}$ is the elastic compliance, and $\dot{N}(\tau)$ is the rate of change of the normal load. The elastic compliance is the inverse of the elastic stiffness $K^{(e)}$

$$C^{(e)} = \frac{d\delta^{(e)}}{dN} = \frac{1}{K^{(e)}} \quad (44)$$

and, from (13) and (14), we have

$$C^{(e)}(N) = \frac{1}{bN + m} \quad (45)$$

$$C^{(e)}(N) = \frac{1}{p} \left(\frac{q+1}{q} N \right)^{-q/(q+1)}. \quad (46)$$

Consequently, the contact radius becomes a function of the normal load history, as expressed by the following equation:

$$a^2(t) = 2R \int_0^t h(t-\tau) \cdot C^{(e)}(N(\tau)) \cdot \dot{N}(\tau) d\tau \quad (47)$$

which is obtained by substituting (43) into (32).

Likewise, the parameter $k(t)$, as expressed in (40), becomes a function of the normal load history. Thus, the convolution integral

$$\int_0^t h(t-\tau) C^{(e)}(N(\tau)) \dot{N}(\tau) d\tau \quad (48)$$

must be considered, and the pressure distribution in (22) also becomes a function of the normal load history.

V. CONSTRUCTION OF LIMIT SURFACES IN VISCOELASTIC CONTACTS

In this section, we consider the construction of the limit surfaces based on the theoretical models of the two cases presented in the preceding sections. The basics of the limit surface construction are presented in Appendix I. The new contribution of this paper is to apply the methodology to study the evolution of limit surfaces in viscoelastic contacts. The evolution of limit surfaces as a time-dependent feature of typical viscoelastic contact interface is presented and discussed.

A. Constant Contact Area

The construction of limit surface in the case of constant contact area is similar to that of an elastic finger in contact. The details can be found in Appendix I. In this case, the integrands in (57) and (58) do not depend on time, and are essentially the same as those of the elastic soft fingers in contact. The two equations suggest that the limit surfaces are evolving, based on the changing normal force $N(t)$, with the shape of each individual limit surface being the same as that of the corresponding limit surface for elastic soft finger at the same normal force N at each time instant. In this case, the shape of the limit surface does not change, but scales proportionally inward as N reduces. If the normal force decreases exponentially, as shown in Fig. 4(a), both the tangential force and the moment will also decrease exponentially.

From (57) and (58), the normalized tangential force and normal moment can be expressed as follows for the construction of the limit surface:

$$\frac{f_t(t)}{\mu N(t)} = \frac{C_k}{\pi} \cdot \int_0^{2\pi} \int_0^1 \frac{\tilde{r}^2 \cos \theta - \tilde{r} \tilde{d}_c}{\sqrt{\tilde{r}^2 + \tilde{d}_c^2 - 2\tilde{r} \tilde{d}_c \cos \theta}} (1 - \tilde{r}^k)^{1/k} d\tilde{r} d\theta \quad (49)$$

$$\frac{m_n(t)}{a\mu N(t)} = \frac{C_k}{\pi} \cdot \int_0^{2\pi} \int_0^1 \frac{\tilde{r}^3 \cos \theta - \tilde{r}^2 \tilde{d}_c}{\sqrt{\tilde{r}^2 + \tilde{d}_c^2 - 2\tilde{r} \tilde{d}_c \cos \theta}} (1 - \tilde{r}^k)^{1/k} d\tilde{r} d\theta \quad (50)$$

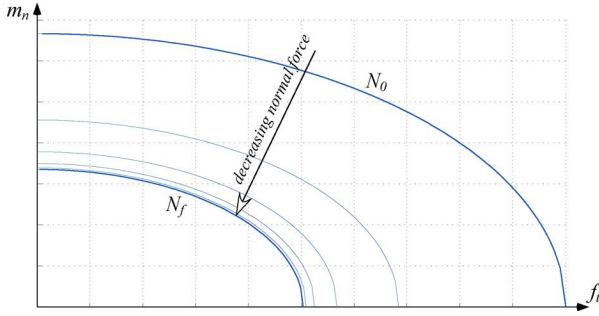


Fig. 9. Sample result of limit surface when the contact area is maintained at constant, in which the constant radius of contact is maintained, resulting in the exponential decrease of the normal force. The limit surface will also scale in proportion to the decay of normal force from N_0 to N_f . The axis f_t is the friction force, and m_n is the moment.

where the normalized coordinates $\tilde{r} = r/a$ and $\tilde{d}_c = d_c/a$ are employed to facilitate the integration.

Since the normal force decreases exponentially and approaches asymptotically a specific value after a certain settling time, as shown in Fig. 4(a), the change of limit surface will follow the same exponential pattern. An example of the evolution of such limit surfaces in the first quadrant is plotted in Fig. 9 by assuming that the pressure distribution and C_k do not change. As illustrated in Fig. 9, the limit surfaces move inward as the normal force $N(t)$ decreases. The exponential decay suggests larger shrinkage initially, while asymptotically approaches a constant surface as $t \rightarrow \infty$.

B. Constant Normal Load

As discussed in Section IV, when the normal contact force is kept constant ($N = N_0$), the normal displacement and radius of contact will gradually relax as time elapses. By employing the equations obtained in Section IV-C and substituting (22) into (53) and (55), we can formulate the normalized force and moment for the viscoelastic contact, similar to (57) and (58), as shown in (51) and (52) at the bottom of the page, where $\tilde{r} = r/a_0$, $\tilde{d}_c = d_c/a_0$, and $\tilde{a} = a/a_0$ are the normalized coordinates, and the parameters $a(t)$, $k(t)$, and $h(t)$ evolve as prescribed in (34), (40), and (41). The limit surface can be constructed using these two equations. However, the evolution of the limit surfaces as time elapses is less intuitive. We present the results of numerical integration of (51) and (52) in Fig. 10. The evolution of the limit surfaces when time elapses can be visualized from the figure as they move outward.

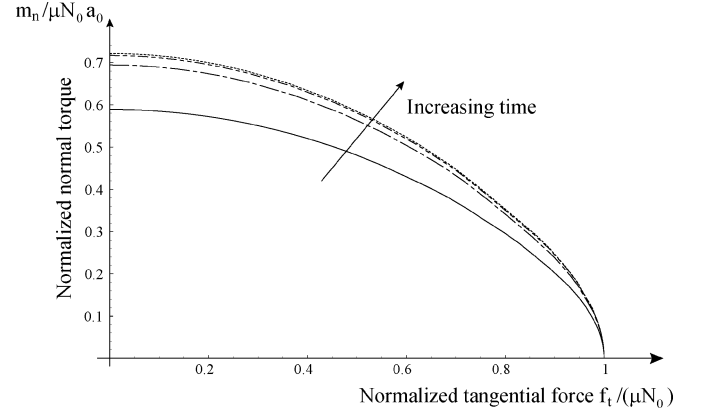


Fig. 10. Evolution of the limit surface as a function of time in the case of constant normal force. The tangential force f_t is normalized with respect to the maximum tangential force at the instant $t = 0$, and the normal moment m_n is normalized with respect to the product $\mu N_0 a_0$, where N_0 is the imposed normal force, a_0 is the radius of contact at the initial instant, and μ is the coefficient of friction.

VI. DISCUSSIONS

The viscoelastic contact models presented in this paper are based on separating the effects of the instantaneous time-independent elastic response and the time-dependent creep or relaxation response. The proposed approach has been demonstrated to be very useful because the model can be obtained by employing elastic modeling previously developed. Recent studies in contact with inclined soft fingers are presented in [30] and [31] using elastic modulus of fingertip and stiffness model in conjunction with geometry. Such studies and previous modeling of soft fingers [3]–[6], [25], [26], [32] address the elastic response in contacts and grasping. Such models give rise to the elastic response as described by $N^{(e)}(\delta)$. Viscoelastic modeling can employ any of such modeling to obtain the elastic response of soft finger and augment it with the temporal response. Using the proposed model, the temporal response and the elastic response can be separated and dealt with independently.

By the nature of the modeling equations, we found that the two important parameters governing the contact (namely, the contact radius a and the profile of pressure distribution k at the contact interface) are not directly related by using the modeling and analytical equations. Therefore, coupling equations are proposed to correlate the two parameters, based on pressure distribution and creep compliance. The former is useful when $2 \leq k \leq 4$, while the latter is more general for all ranges of k values. It is noted that both equations satisfy the equilibrium

$$\frac{f_t(t)}{\mu N_0} = \frac{1}{\pi} \int_0^{2\pi} \int_0^{\tilde{a}(t)} \frac{\tilde{r}^2 \cos \theta - \tilde{r} \tilde{d}_c}{\sqrt{\tilde{r}^2 + \tilde{d}_c^2 - 2\tilde{r} \tilde{d}_c \cos \theta}} \frac{C_{k(t)}}{\tilde{a}(t)^2} \left[1 - \left(\frac{\tilde{r}}{\tilde{a}(t)} \right)^{k(t)} \right]^{1/k(t)} d\tilde{r} d\theta \quad (51)$$

$$\frac{m_n(t)}{\mu N_0 a_0} = \frac{1}{\pi} \int_0^{2\pi} \int_0^{\tilde{a}(t)} \frac{\tilde{r}^3 - \tilde{r}^2 \tilde{d}_c \cos \theta}{\sqrt{\tilde{r}^2 + \tilde{d}_c^2 - 2\tilde{r} \tilde{d}_c \cos \theta}} \frac{C_{k(t)}}{\tilde{a}(t)^2} \left[1 - \left(\frac{\tilde{r}}{\tilde{a}(t)} \right)^{k(t)} \right]^{1/k(t)} d\tilde{r} d\theta \quad (52)$$

equation and physical coherent behavior of viscoelastic contacts. The reduced creep compliance $h(t)$ is introduced, which depends only on the viscous properties of the material, for the modeling of the creep behavior at the contact interface. By using this function, the evolution of both the pressure distribution, depending on k , and the contact radius a , can be obtained. A noteworthy conclusion is that both variables k and a are related to the same function $h(t)$ and, moreover, they depend on the geometry.

Two cases are studied with respective modeling equations and presented with the evolution of limit surfaces as time elapses. The two cases are constant contact area and constant normal force. In the former case with constant contact deformation, and thus constant contact radius, it is found that the relaxation and reduction of normal force makes the limit surfaces shrinking asymptotically, resulting in a less stable grasp due to its viscoelastic behavior. This is illustrated in Fig. 9. In the latter case with constant normal force, the contact area increases due to creep. As a result of the growth of the contact area while maintaining the same normal force, the limit surface expands, enhancing the stability of grasping at the contact interface. This is illustrated in Fig. 10, in which the limit surfaces change shapes (do not scale proportionally) because the maximum tangential force (when $m_n = 0$) depends only on the normal load and the friction coefficient, but is not affected by the viscoelastic phenomena, while, on the other hand, the maximum normal moment increases because the contact area grows and the pressure distribution becomes more uniform. Since the normal to the limit surface represents the instantaneous direction of sliding, this result also has profound implications on the grasping and manipulation using viscoelastic fingertips.

In typical grasping and manipulation tasks, normal force is maintained and controlled instead of the contact deformation. Based on the preceding discussions, it is concluded that such grasp with viscoelastic contacts will become more stable as time elapses, due to the time-dependent nature of viscoelasticity (see Fig. 10). While this is intuitive, the viscoelastic modeling provided in this paper explains such an intuition. Thus, robotic fingers provided with pads that exhibit prominent viscoelastic behavior are more advantageous, because the relaxation phenomena arise and a progressive growth of contact area under the same grasping forces takes place, resulting in expanded limit surface and increased stability against sliding at the contact interface.

Nonetheless, we note that manipulation without detailed contact modeling was presented in various work [33]–[37] pertaining to sensorless manipulation and distributed manipulation. This paper presents a manageable modeling of viscoelastic contact that can be incorporated into the analysis of contacts to address the elastic and temporal responses of contacts. Such modeling, when realized, will enhance the ability in manipulation by using the characteristics of contacts.

VII. CONCLUSION

In this paper, we study the influence of the time-dependent relaxation nature of viscoelastic fingers in contact interface for robotic grasping and manipulation. Because the time-dependent nature of the viscoelastic finger is generally independent of the elastic response, the principle of separation between the elastic

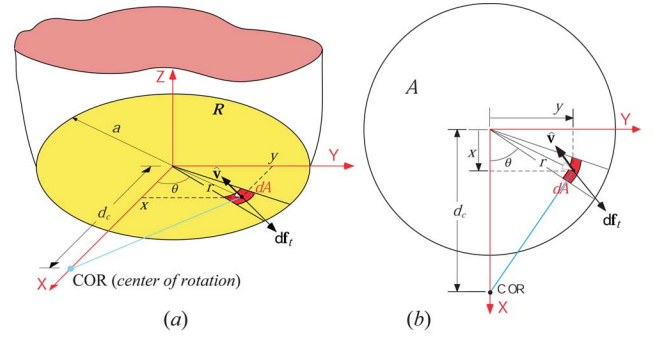


Fig. 11. Schematic showing the COR and the integration of force and moment on a contact patch [6], [10]. (a) 3-D perspective view. (b) Element for integration on the XY plane along the area A . The COR is chosen to move along the X axis to span the entire limit surface.

and temporal responses is employed for modeling and analysis. It is deduced when analyzing the theoretical model that the two important parameters describing viscoelastic contacts, namely, the contact radius a and profile of pressure distribution k , can be correlated based on the physical behavior of the contact interface and the properties of the materials. Two different cases applicable to robotic grasping or manipulation are discussed: constant contact area and constant normal force. The evolution of friction limit surfaces and pressure distributions of the two cases were studied and presented. It is found that the control of the grasp forces (case 2) when using viscoelastic contacts is more profitable, because it enhances the stability of grasping through the expansion of limit surface as time elapses. The analysis proposed in this paper is useful in the synthesis and analysis of viscoelastic fingers/feet and contact interface.

APPENDIX I

EQUATIONS OF TANGENTIAL FORCE AND MOMENT IN CONSTRUCTING LIMIT SURFACE FOR VISCOELASTIC CONTACT

The equations of tangential force and moment for viscoelastic fingers follow the same derivation based on the scanning of instantaneous center of rotation (COR),⁵ initially proposed in [10]. The tangential force $\mathbf{f}_t$ over the entire contact area is the sum of the friction forces $d\mathbf{f}_t$, as shown in Fig. 11, governed by the Coulomb's law on the infinitesimal element dA [2], [6]

$$\mathbf{f}_t = \int_A d\mathbf{f}_t = - \int_A \mu \hat{\mathbf{v}} p(r) dA. \quad (53)$$

The vector $\hat{\mathbf{v}}$ in (53) represents the unit vector along the direction of the velocity (opposite to the direction of $d\mathbf{f}_t$) with respect to the COR, as shown in Fig. 11, μ is the coefficient of friction, and $p(r)$ is the pressure on the infinitesimal contact patch. Since we choose the coordinate frame such that the COR is along the X axis, it can be deduced that $f_x = 0$ due to symmetry. Thus, the force vector in (53) can be expressed as a scalar $f_t = f_y$. The y -component of the unit vector can be expressed as [6]

$$\hat{v}_y = \frac{r \cos \theta - d_c}{\sqrt{r^2 + d_c^2 - 2r d_c \cos \theta}} \quad (54)$$

⁵That is, by varying the distance d_c in Fig. 11 with COR moving along the X axis.

where r and d_c are the distances shown in Fig. 11, and θ is the angle shown.

Similarly, the moment about the normal to the contact area in the Z direction is

$$\mathbf{m}_n = - \int_A \mu [\mathbf{r} \times \hat{\mathbf{v}}] p(r) dA. \quad (55)$$

Since $\mathbf{m}_n$ is along the Z direction, the scalar notation of m_n representing the magnitude will be used. The magnitude of the cross product in (55) becomes [6]

$$|\mathbf{r} \times \hat{\mathbf{v}}| = \frac{r^2 - r d_c \cos \theta}{\sqrt{r^2 + d_c^2 - 2r d_c \cos \theta}}. \quad (56)$$

Note that (53) and (55) remain the same for viscoelastic contact, except that the terms f_t and m_n are now functions of time and depend on the nature of the load, because the pressure p is a function of time. By combining terms from (53) to (56), the following equations are derived:

$$f_t = \frac{\mu N}{\pi} \cdot \int_0^{2\pi} \int_0^a \frac{r^2 \cos \theta - r d_c}{\sqrt{r^2 + d_c^2 - 2r d_c \cos \theta}} \frac{C_k}{a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} dr d\theta \quad (57)$$

$$m_n = \frac{\mu N}{\pi} \cdot \int_0^{2\pi} \int_0^a \frac{r^3 - r^2 d_c \cos \theta}{\sqrt{r^2 + d_c^2 - 2r d_c \cos \theta}} \frac{C_k}{a^2} \left[1 - \left(\frac{r}{a} \right)^k \right]^{1/k} dr d\theta. \quad (58)$$

APPENDIX II

LS BEST FIT FOR THE COUPLING (26)

From Section IV-B, the radius a is assumed to be proportional to the inverse of the square root of the coefficient C_k , as in (25). Furthermore, we assume that $1/\sqrt{C_k}$ is related to the parameter k in a typical exponentially decaying form of

$$\frac{1}{\sqrt{C_k}} = 1 - \psi e^{-\xi k}.$$

Thus, it can be written that

$$\frac{a}{a_0} = c(1 - \psi e^{-\xi k}) \quad (59)$$

where a/a_0 is the normalized radius of contact, a_0 is the initial contact radius at $t = 0$ with $k = 2$, and c, ψ, ξ are constants to be determined. LS fit of such a transcendent equation can be formulated [26] as follows. First of all, to fit data $y = 1/\sqrt{C_k}$ with $(1 - \psi e^{-\xi k})$, we use the following equation:

$$y_i = 1 - \psi e^{-\xi k_i} \implies \log(1 - y_i) = \log \psi - \xi k_i. \quad (60)$$

Formulating the above equations in a matrix form for the LS fit, we have

$$\mathbf{A} \mathbf{x} = \mathbf{b} \quad (61)$$

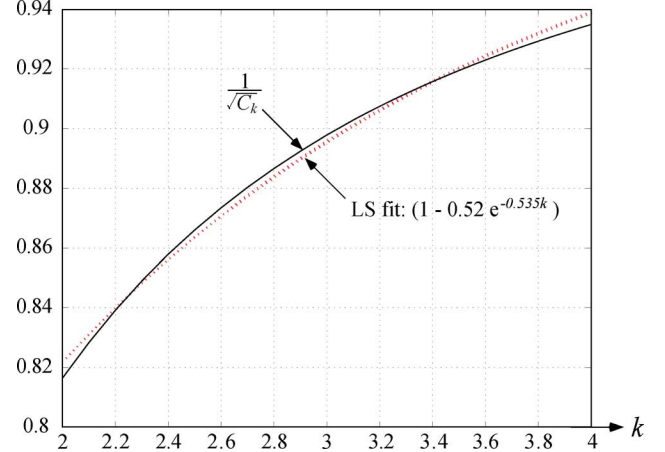


Fig. 12. Comparison of the theoretical values of $1/\sqrt{C_k}$ versus the weighted LS fit of (64).

where

$$\mathbf{A} = \begin{bmatrix} -k_1 & 1 \\ -k_2 & 1 \\ \vdots & \vdots \end{bmatrix} \mathbf{x} = \begin{bmatrix} \xi \\ \log \psi \end{bmatrix} \mathbf{b} = \begin{bmatrix} \log(1 - y_1) \\ \log(1 - y_2) \\ \vdots \end{bmatrix}. \quad (62)$$

The LS best fit of the solution $\mathbf{x}$ in (61) containing ξ and ψ is given in (63) involving the generalized inverse [26], denoted by the superscript “ $^{+}$ ”

$$\mathbf{x} = \mathbf{A}^+ \mathbf{b} \text{ or } \mathbf{x} = (\mathbf{W}\mathbf{A})^+ (\mathbf{W}\mathbf{b}) \quad (63)$$

where $\mathbf{W}$ is a weighting matrix, typically chosen as $e^{\mathbf{b}}$ to scale the logarithmic algorithm in (60).

A table of C_k values with selected k is listed in Table II. Applying (63) for data of $2 \leq k \leq 4$ with weighting, we can determine

$$\frac{1}{\sqrt{C_k}} \cong 1 - 0.52 e^{-0.535k}. \quad (64)$$

Combining (64) with (25), we can derive

$$\frac{a}{a_0} \cong c(1 - 0.52 e^{-0.535k}) \quad (65)$$

where $c = 1.217$ in order to render a normalized radius of contact. Equation (65) has been found to work well when $2 \leq k \leq 4$, as shown in Fig. 12. Table II lists a few sample values of k between 2 and 4 with corresponding values of a/a_0 . Note that when a larger range of k is adopted, the LS fit may render slightly different values of the coefficients and exponent in (65).

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Paolo Tiezzi received the university degree and the Ph.D. degree from the University of Bologna, Bologna, Italy, in 2002 and 2006, respectively.

He is currently with the Department of Mechanical Engineering (DIEM) at the University of Bologna. He conducts research in the areas of robotics and automation, in particular, soft and viscoelastic contact interface for robotic manipulation, anthropomorphic robotic hands, and linear synchronous brushless motors for application on automatic machines.



Imin Kao (S'88–M'90) received the B.S. degree from National Chung Hsing University, Taichung, Taiwan, R.O.C., in 1981, and the M.S. and Ph.D. degrees from Stanford University, Stanford, CA, in 1986 and 1990, respectively.

A professor of the Department of Mechanical Engineering at the State University of New York (SUNY) at Stony Brook, he is also the founding Faculty Director of the Information and Technology Studies Undergraduate College (ITS College) at Stony Brook. Being the Director of the Manufacturing Automation Laboratory (MAL) at Stony Brook, he conducts research in the areas of robotics and manufacturing automation, MEMS, intelligent contact interface, stiffness control, soft contacts in grasping and dextrous manipulation, wafer manufacturing, and wafer slicing using wiresaw.

Dr. Kao served as an Associate Editor of the IEEE TRANSACTIONS ON ROBOTICS AND AUTOMATION, as well as the *International Journal of Advanced Manufacturing Systems*. He is a member of the ASME.